Sections and retractions

Soham Saha

https://soham-saha.github.io

https://soham-saha.codeberg.page

- Given a surjective function $f : A \to B$, each of the fibers over elements of B are inhabited. Therefore, we can define a function $g : B \to A$ which picks out one element from each fiber; i.e., $g(b) = x$ for some $x \in f^{-1}(b)$. Such a g is called a **section** of f, and it satisfied $fg = \mathrm{id}_B$.
- Given an injective map $g : B \to A$, a map $f : A \to B$ which respects the bijection between B and $\mathrm{im}(g)$ is called a **retraction** of g, i.e., $g(x) = y \Rightarrow f(y) = x$. Such an f satisfies $fg = \mathrm{id}_B$.
- If we have $g : B \to A$ and $f : A \to B$, such that $fg = \mathrm{id}_B$, then g is forced to be injective, while f is forced to be surjective. In this case, g is called a section of f, and f is called a retract of g. Also in this case, B is called the retract of A, since we can "retract" from A to B using f.

